

A Minor Project Report

on

SnapTalk: Image Captioning and Description in Gujarati

by

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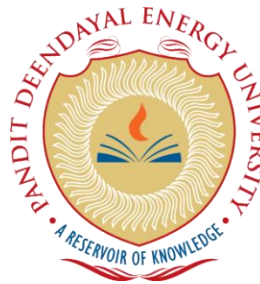
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Submitted to



Department of Computer Science and Engineering,

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2024

CERTIFICATE

This is to certify that the seminar report entitled “**SnapTalk: Image captioning and description in Gujarati**” submitted by **Ansh Raiyani & Dev Patel** has been conducted under the supervision of **Dr. Amitava Choudhury, Assistant Professor**, and is hereby approved for the partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in the Department of **Computer Science and Engineering** at Pandit Deendayal Energy University, Gandhinagar. This work is original and has not been submitted to any other institution for the award of any degree.

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Acknowledgement

We would like to thank each and every one of those who stood by us during the entire span of our project ‘AI Powered Image Captioning System for the Gujarati Language’ which we are glad to say was successful.

To begin with, we are thankful to our mentor, Dr. Amitava Choudhury for his support in every step. His experience as well as motivation were proudly utilized in order to gain more insight regarding advanced AI methodologies along with various fine-tuning techniques. The ideas as a whole have improved the level of our work and encouraged us to examine more opportunities to advance AI in the realm of low-resource languages.

The appreciation goes to the faculty and fellow students of Pandit Deendayal Energy University who gave feedback and presented critiques which were extremely useful. Their and much more were extremely efficient in defining our strategy and enhancing the cogency and applicability of our arguments.

This project could only have been achieved thanks to the commitment, hard work and enthusiasm of every single member of our team. We would like to thank anyone who played a role in this endeavour and made it worthwhile. A special thanks to all of you for directing us into making this project a triumph.

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Declaration

I hereby declare that the seminar report entitled **SnapTalk: Image captioning and description in Gujarati** is the result of my own work and has been written by me. This report has not utilized any language model or natural language processing artificial intelligence tools for the creation or generation of content, including the literature survey. The use of any such artificial intelligence-based tools was strictly confined to the polishing of content, spellchecking, and grammar correction after the initial draft of the report was completed. No part of this report has been directly sourced from the output of such tools for the final submission.

This declaration is to affirm that the work presented in this report is genuinely conducted by me and to the best of my knowledge, it is original.

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List of Tools Used for the Report with Purpose: ChatGPT (Correcting Grammar)

Abstract

The current project presents the image captioning system for Gujarati, a language which has been marginalized in terms of an AI-based solution. Such application uses the BLIP (Bootstrapping Language-Image Pre-training) architecture, which offers excellent multi-modality performance as a language vision unified understanding and generation framework. It also ensures that the system generated captions that are contextually and linguistically appropriate for the Gujarati language by employing modifications, guided by an annotated data set specifically designed for the Gujarati language [1][8]. The project's approach aims is mindful of the low resource languages and its challenges such as limited data coverage and language heterogeneity, while also catering to the rising demand for regional AI based answers [2].

Implementing a comprehensive approach, the dataset construction, transfer learning and application of evaluation metrics decisively to precede targeting and evaluation of language beauty along the lines of BLEU and ROUGE for geometric correctness [3][4]. A number of sophisticated solutions were applied to avoid catastrophic forgetting when fine-tuning the model, which would alter its over-generalization capabilities because of adaptation to Gujarati language [12]. The finished product incorporates a simple but elegant UI enabling users to upload picture and obtain Gujarati caption for the image.

Apart from automated caption generation, the system offers user feedback capabilities, allowing the model to improve over time. Future iterations will aim at including a user image upload feature that will help in the growth of a dataset and improve the efficiency of models through feedback loops. Such an approach, driven by user feedback, helps bring about improvement in the captioning quality and captioning flexibility [9][10].

In addressing linguistic inclusion, this project makes an important step in the evolution of AI as it goes on to show a model of AI system development that is robust for enabling low-resource language AI. It not only benefits the Gujarati community but also provides a way forward on how the AI expansion can be done within other under-resourced languages.

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List of Abbreviations

- DL – Deep Learning
- NLP – Natural Language Processing
- AI – Artificial Intelligence
- ML – Machine Learning
- BLIP – Bootstrapping Language Image-Image Pretraining
- BLEU – Bilingual Evaluation
- ROUGE – Recall-Oriented Understudy for Gisting Evaluation
- CIDEr – Consensus-based Image Description Evaluation
- DNN – Deep Neural Network
- CNN – Convolutional Neural Network
- RNN – Recurrent Neural Network
- VGG16 – Visual Geometry Group 16
- LSTM – Long Short-Term Memory
- MS-COCO – Microsoft Common Objects in Context
- ViT – Vision Transformer
- GPT2 – Generative Pretrained Transformer 2
- API – Application Programmable Interface
- ADAM – Adaptive Moment Estimation
- TTS – Text to voice

List of Nomenclature

- React Native – Javascript Framework for Mobile Application Development
- FastAPI – Python library for API development
- Docker – Platform for application containerization
- CNN – A neural network architecture that is able to identify certain patterns present in images such as borders, textures, and objects automatically.
- RNN – A branch of artificial neural networks that is effective for sequential data by usage of its internal memory for time based information.
- DNN – A neural network architecture that has several hidden layers with the purpose of modeling complex, non-linear relationships in data.
- LSTM – A type of recurrent neural network that is capable of learning long term dependencies because it solves the problem of the disappearing gradient.
- VGG16 – This is a deep convolutional network with 16 layers that is recognized for successfully performing image classification and extracting useful information about images.
- EfficientNet – Different models of CNN architectures that aim at achieving great accuracy and high performance by balance in depth, width and resolution.
- DevOps – A set of practices that combine software development and IT operations to improve collaboration, automate workflows, and accelerate the delivery of high-quality software.
- MLOps – A practice that applies DevOps principles to machine learning, focusing on automating and streamlining the deployment, monitoring, and maintenance of ML models in production.

Chapter – 1: Introduction

Deep learning (DL) and natural language processing (NLP) are two rapidly evolving areas of study that have applications in several domains, such as in image captioning systems which enable computers to create meaningful text for images. However, this development is often only seen in high-resource languages like the English, while regional languages such as the Gujarati remain largely untapped. Gujarati which is spoken by millions across the globe, suffers from the absence of any substantial AI tools or datasets, which creates barriers to fully employ the AI's capabilities in this language. Not only does this gap impede digital equity, but it also stifles the growth of expressive industries that are dependent on AI-enabled technologies.[1]

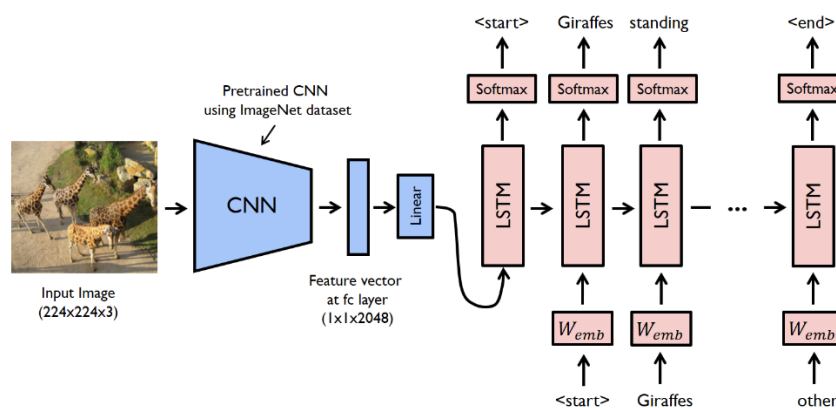


Figure 1 – General Image Captioning Process

Note that in the past few years, the work on integrating vision and language has improved the way images are captioned within a paradigm. Such models as BLIP have revealed that pretrained vision-language models produce great results in English captioning tasks. The situation is, however, different with low resource languages such as Gujarati where the lack of any trained datasets and the diversity of the pretraining context limits usability. Closing this gap would improve the AI reach and extend the scope for the digital inclusion of less served language communities.[4]

In dealing with this difficulty, the Snaptalk project attempts to improve the blip image captioning large model for the Gujarati language [4]. Also, based on a vision-language pretraining, this model appears to be well suited for image captioning tasks. The focus of the project is to design an Indian dataset that is of good quality that is composed of images that are culturally appropriate and Gujarati captions that are audio described. It has been established that extending the BLIP model to this data results in accurate and culture focused captions, and hence the task is achievable in AI systems for low resource languages. For creating the dataset we used several techniques including web searching for images on sites such as Flickr and taking pictures ourselves. This made it possible to acquire a wide range of visual materials including mundane and traditional cultural items. To guarantee translation accuracy and cultural relevance, each image's captions, which were initially in English, were also translated and annotated in Gujarati [2]. This dataset is crucial for fine-tuning the BLIP model as there are no Gujarati image-caption pairs available in the existing repositories.

The fine-tuning process involved image resizing and captioning in Gujarati [1]. After this measure was taken, the BLIP model incorporated continual learning methods and preserved its general image tagging skills, topographies related to the region. Dual approach research which included quantitative analysis of BLEU scores [3], as well as perception analysis of markers and similarities of the caption to the culture was conducted on the model. The results of these evaluations showed that the model was capable of generating appropriate captions in the language of Gujarati, thus, indicating opportunities for AI remedies to move the low-resource language communities.

This research targets further wider goals of making AI interventions for all. The work of the project in focus to describe the language of Gujarati helps to widen the digital gap and increasing the technological penetration into the areas which are not represented [4]. The findings of Snaptalk show the necessity for building language-centered datasets and fine-tuning pre-trained AI models for regional languages and culture. Thus, the work does not only contribute to the advancement of the technology of image captioning but as well provides a guideline for other researches attempting to integrate AI in the development of the language of the region.

Chapter – 2: Literature Survey

Image description is complex task that sit between computer graphics and linguistic fields whereby text is produced for an image provided. Dependence on an image with a caption made it easier for the authors [5] to devise an image captioning DNN based framework. With their effort, caption generation was considerably enhanced in terms of the coherence and relevance. Their model effectively employed visual features extracted from CNN and combined them with RNN for sequence generation, and reported accuracy of BLEU-1, BLEU-2, BLEU-3 and BLEU-4 as 0.701, 0.528, 0.404 and 0.308 respectively when applied on MS-COCO dataset.

In recent years, a lot of attention has been devoted to surmounting the key challenges posed by low-resource languages. One such attempt includes the development of the Assamese language captioning model [6], which adapted available models to develop captions for a low-resource dataset. They used VGG16 and EfficientNet as their encoder model. The combined BLEU was 0.2134 and 0.2389, respectively. While the ROUGE scores came out to be 0.0778 and 0.0889, respectively. Correspondingly, researchers built a hybrid model with CNN and LSTM networks for Hindi captioning and showed better performance by customizing the system to the designated language [7].

The BLIP [8] framework was able to bridge the gap between visual and language understanding tasks. It combines vision and text transformers, which can be put to use for a wide array of tasks with much better performance than other conventional techniques, including image captioning. BLIP achieved a CIDEr score of 134.9 and a BLEU-4 score of 36.9, surpassing previous models in generating fluent and accurate captions. In this model, there is a form of dependency where pre-training on large-scale datasets ensures much better fixation in the low resource setting of downstream tasks.

Efficiency in resource utilization has also been a focal area. Language resource-efficient learning approaches have been explored to optimize training for captioning tasks, reducing computational demands without sacrificing performance [9]. Further, techniques like cycle consistency have been employed to improve captioning for low-resource languages, ensuring generated captions remain semantically aligned with the image content [10].

The advantages of Transformer architectures in the simultaneous processing of both vision and text inputs have made them interesting to many researchers. Vision/text transformer-based image captioning models have achieved a CIDEr score of 141.7 on the MS-COCO datasets, creating sophisticated captions thanks to the fused visual and text components [11].

Dealing with continual learning is another issue in image captioning, where the models are required to incorporate new data while not forgetting the previously learned tasks. However, ContCap and other approaches are scalable to continual image captioning as they avoid catastrophic forgetting with new regularization methods [12, 13]. These methods are important when new datasets become available and need to be added to the model's training.

Finally, in their panorama surveys of the deep learning methodologies in image captioning the evolution of the field's development is traced, indicating its strengths and weaknesses and proposing further research areas [14].

Chapter – 3: Methodology

The methodology presents the theoretical framework and strategies employed in coming with an efficient Gujarati image captioning system. It discusses reasons for choosing pretrained models, assessment approaches and factors that make the use of modern advanced technologies possible in a low resource language such as Gujarati.

1. Vision-Language Integration

Image Captioning is a multi-layered task since it requires both computer vision and natural language generation creating the need for robust models that would enable both to work together. Transformers, especially vision-text architectures such as BLIP, have worked well in catching up with picture understanding and the generation of text. This can be due to their nature of being able to perform use both the visual information and the text information at the same time, due to usage of attention mechanisms to target the important aspects and relations between the two modalities [1]. The Centre for Bus and Traffic Studies (2017) states that the BLIP model is suitable since it integrates Computer Vision and Natural Language Processing. Since a Vision Transformer (ViT) was designed to perform the task of image understanding whilst a text transformer was able to perform the task of image captioning. This made the design optimal since it was performing two tasks efficiently [8].

2. Challenges in Low-Resource Languages

Languages like Gujarati, spoken by millions around the world, are often overlooked in AI systems. Custom datasets are essential to cater for cultural and linguistic differences to AI solutions. For example, the Indian language which has been used widely has not been able to leverage current systems effectively, owing to a shortage of datasets that are annotated and domain-specific [2].

3. Fine-Tuning for Regional Contexts

This process of fine-tuning enables specialized domain tasks, or a domain few shot, to be performed with a fine-tuned model while mitigating catastrophic interference. In the case of BLIP, this was possible due to the application of regularization and dual encoders allowing the model to learn other tasks while also image captioning [12]. Given such characteristics, BLIP can easily be adapted to low-resource contexts such as the Gujarati language.

4. Text Translation

We leveraged a specialized BERT model in order to convert captions produced in English by the BLIP algorithm to captions in Gujarati. This model was developed to work with publicly obtainable Gujarati datasets, and specifically geared for working in machine translation environments. Such language barrier was addressed by adopting this translation model which acted on BLIP's English captions to create quality captioning in Gujarati that is both linguistically and culturally appropriate. It also reduces the divide between a resource abundant model and a resource scarce target language, thus making it easier to image caption for the Gujarati language speakers.

5. Evaluation Metrics.

Standard evaluation metrics like BLEU (Bilingual Evaluation Understudy) and ROUGE (Recall-Oriented Understudy for Gisting Evaluation) were employed in measuring the models outputs. BLEU measures the n-gram precision score between generated captions and reference captions, while the ROUGE measures the overlapping sequences recall based on textual information, thus enabling for a more all rounded assessment on the dependence and information value of the captions [3] [4].

6. Cultural Relevance and Dataset Design.

A significant part of the methodology used was the design of an appropriate dataset with cultural context. Images with Indian settings such as structures, events, dressing styles and cuisine, were labelled in the Gujarati language to meet the language and cultural context requirement. This approach was able to resolve the problem of the unavailability of datasets for Gujarati image captioning [6] by maintaining quality captions through manual annotation.

7. Rationale for Model Selection.

Early experiments with ViT-GPT2 indicated problems of catastrophic forgetting and generalizability on a reduced set. These issues created an opportunity for a stronger model such as BLIP that performs well due to its pretraining on broad data sets and strong synchrony between vision and language in low resource tasks [11] [13].

8. Future Implications

This project not only advances AI technology for Gujarati speaking users, but it also serves as a stepping stone for the future development of other low resource languages. The methodology confirms that extension of the vision language pretrained models to be culturally contextualized is made possible, which enhances digital inclusivity and minimizes technological inequalities [9] [10].

In subsequent versions, we will include a user feedback cycle and allow the app to incorporate future improvements automatically. Users will ultimately take images through the app which will be uploaded on the server and will later be incorporated into the model for retraining purposes with the aim of improving the overall quality of the captions. Also, the app will have a ‘suggestion box’ where the users can send messages to propose more appropriate captions or point out wrong captions and therefore providing useful information for subsequent model improvements.

These improvements will make the model incorporate user suggestions and update later versions with the intention of making the captions more accurate and relevant with the context. This participatory approach of users engaging with the AI offers opportunities for the users to directly assist in the improvement of the models while providing a better experience.

Chapter – 4: Procedure

The procedure describes the steps undertaken to adapt the BLIP model aimed at the generation of English captions for images pertaining to the Indian context specifically, the work undertaken in this regard includes data collection, its preparation, model tuning, and its assessment.

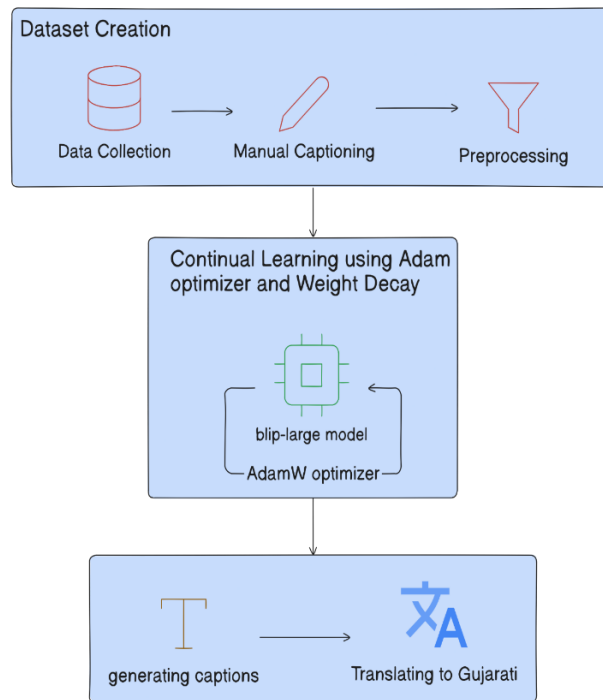


Figure 2 - Work Flow Diagram

1. Data Collection

In order to meet the aim of generating captions that align with Indian culture, the authors used a varied dataset that they collected from a variety of sources. The authors obtained most of the images from Flickr, which complimented their pictorial idiom of various scenes. Also, some images were taken from personal handheld devices to add more dimensions to the images contained in the database. The custom image dataset was about more than 1,800 images which comprised different categories such as monuments, festivals, statues of gods, food images, prominent places and rural landscapes of Gujarat. The picture in this case was complemented with a corresponding English caption, which was even more accurate and precisely described what was shown in the picture. Manual annotation

of images was necessary since it allowed them to adjust the model in a way that it produced captions that were specific and appropriate within the context.

2. Data Pre-processing

Data pre-processing was a crucial step in preparing the dataset for fine-tuning the BLIP model. A custom data loader was created using the PyTorch library to handle the dataset efficiently. This loader paired each image with its corresponding caption, adhering to the specific data structure required by the BLIP model. Since BLIP relies on a particular arrangement of image-caption pairs for effective training, this customization was essential to ensure compatibility and optimize the training process.

3. Model Selection

During the early stages of the project, the generating of image captions was performed using ViT-GPT2 model. Although this model was able to perform the task of joining both text and image modals, this model had serious drawbacks. One of the most critical was catastrophic forgetting, where performance of the model fell off unexpectedly when fine-tuned on our data. The problem was aggravated by the fact that the size of our dataset was relatively small, which made the generalizability of the model weak and ineffective. Thus, ViT-GPT2 was not appropriate for the particular case of our work.

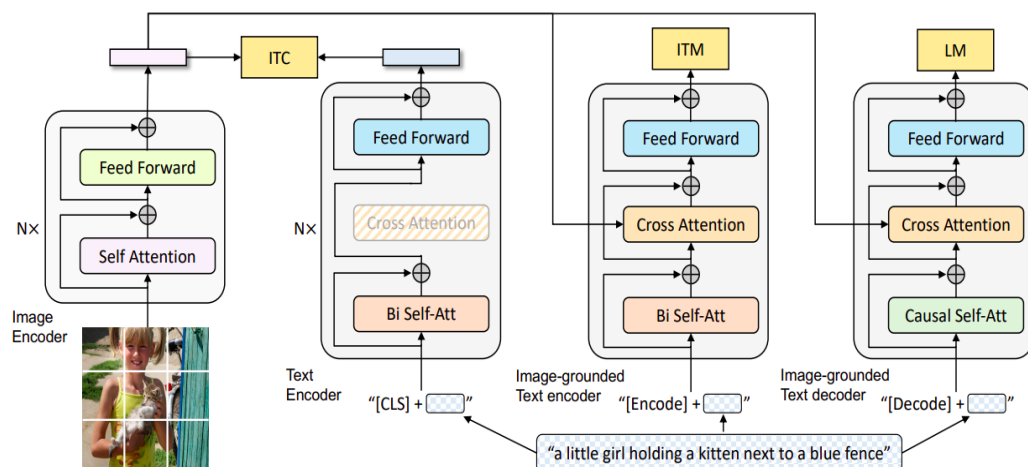


Figure 3 - BLIP model architecture

In order to overcome these difficulties, the BLIP (Bootstrapped Language-Image Pre-training) large model was chosen to tackle the tasks at hand. Especially the large version of BLIP solves the problems of such models as ViT-GPT2 using a Vision Transformer (ViT) for visual processing and a strong Transformer architecture for textual description generation. What makes this model unique is its extensive training on large and varied datasets making it more generalized and resistant to being fine tuned to a single domain.

Reasons for BLIP-Large to Have Succeeded:

- a. **Better Management of Catastrophic Forgetting:** As a result of BLIP's design and training procedures, it is able to preserve its learned knowledge regarding tasks even after being trained further on tasks which are more specific but consist of minimal data sources.
- b. **Improved Vision-Language Synchronization:** The use of Dual Encoder which incorporates image and text representations in a better way thus increasing the quality of the caption produced by the model.
- c. **Achievement on Under-Resourced Domains:** Large cross-lingual representations are beneficial for BLIP which allows it to perform well on cross linguistic recurring tasks even if they are low resource tasks like, creating English captions for Indian content based images.

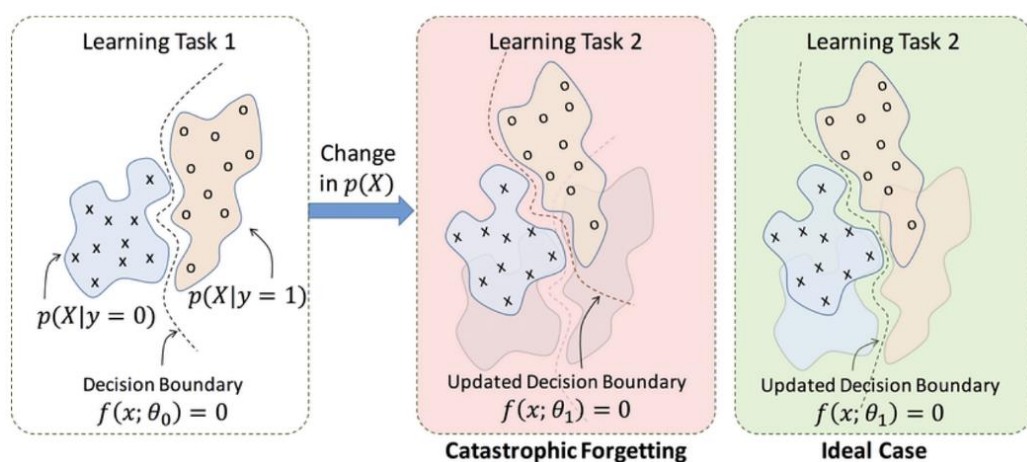


Figure 4 - Catastrophic Forgetting

4. Fine-Tuning

Fine-tuning refers to taking a model which has already been trained on a large dataset, also known as a pre-trained model, and further training this model's parameters for the targeted task at hand or the new dataset. In our case, BLIP-large model was fine-tuned using our dataset so as to cater for generation of text captions specific to images taken in India.

Fine-tuning was necessary as the BLIP model was pre-trained on a large dataset but never seen images on our dataset. With fine-tuning, the model was able to associate text to pictures such as Indian festivals and places such as that of Gujarat with words describing them. This meant the model was able to produce captions that were not only relevant but also descriptive considering the rich culture depicted in the pictures.

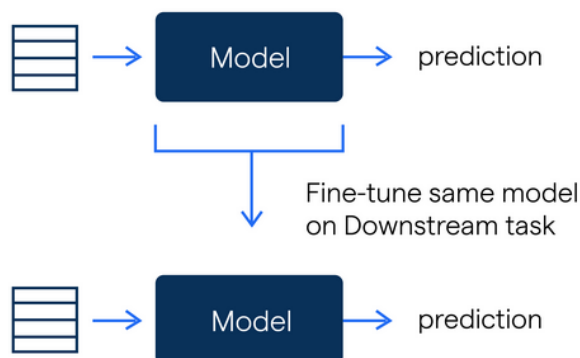


Figure 5 - Fine Tuning

In the case of fine-tuning, the manually captioned dataset has been used to change the weights of the model. AdamW optimizer was used for 10 epochs with the learning rate of $2e-6$. In this way, the model was able to learn the specifics of the task and this as well led to enhancement of the model performance in the specific domain.

5. Evaluation

To assess the efficacy of the fine-tuned BLIP-Large model, a validation dataset comprising 500 images was generated by applying the same approach used in the training dataset. The purpose of this dataset was to evaluate the model's performance in producing captions that were accurate and appropriate for the context. The validation images were accompanied by captions that had been elaborated in order to provide a point of reference for the evaluation.

The results were examined using two metrics that are frequently employed in the task of image captioning:

- a. BLEU Score (Bilingual Evaluation Understudy Score): It assesses n-gram overlap (unigrams, bigrams etc.) between generated captions and reference captions with an emphasis on n-gram precision.
- b. ROUGE (Recall-Oriented Understudy for Gisting Evaluation): ROUGE evaluates the recall of n-grams present in the captions created and those supporting the image and thus indicates how much of informative parts in the images the model is able to comprehend.

6. Deployment

In the deployment of the image captioning model, a application-based interface was developed using React Native, enabling people to use a fine-tuned BLIP-Large model and generate captions for uploaded images. The model was deployed on the cloud for scalability and fault tolerance. An API based on Flask was developed for the model that receives an image soup and provides the soup along with the caption regarding the image.

For convenience, the images were uploaded to the backend first, then pre-processed and forwarded to the model for caption generation. After the object Recognition and caption generation process, the captions were pushed back to the users via the API.

Chapter – 5: Results and Analysis

The purpose of this chapter is to provide the main results of the Gujarati image captioning project with emphasis on both the number of and the nature of achievements. It also demonstrates the usefulness of the model by presenting evaluation measures, visual inferences, and practical applications.

1. Loss vs Epoch curve

During the training, a graph of the loss function measure with respect to the number of epochs was plotted to examine the progress of the model in learning. The plot demonstrated a continuous decrease in the value of the loss, which suggests that the model was successfully trained to reduce the amount of errors made while training. This visualization confirmed the selection of hyperparameters and the goals of the fine-tuning stage.

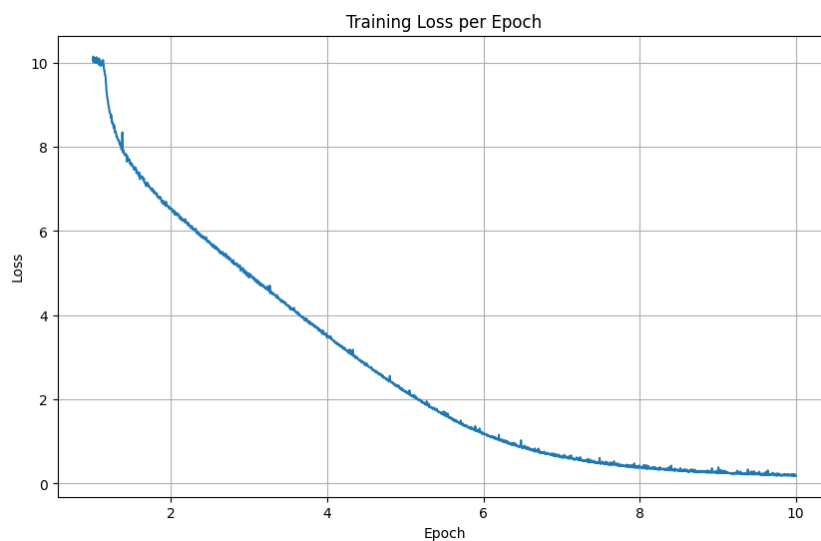


Figure 6 - Loss-Epoch Curve

2. Evaluation Metrics

To examine the performance of the fine-tuned BLIP model, standard evaluation metrics were adopted such as BLEU and ROUGE, which allow for a more rounded measurement of the model's effectiveness in caption generation.

BLEU Scores		
1-gram	2-grams	3-grams
0.533	0.470	0.324

Table 1 - BLEU Scores

ROUGE Scores		
1-gram	2-grams	LCS
0.610	0.320	0.375

Table 2 - ROUGE Scores

With these findings, the model has the ability of capturing the important regions and generating proper captions in the Gujarati language. The captions display a high degree of lexical overlap with BLEU-1 and ROUGE-1 scores, and the model's higher order captioning abilities can be assessed with the upper scores of the two metrics focus on context capturing.

3. Inference

In order to assess the generalizability of the model, an inference chart was made with absent images. It is also worth pointing out that the captions produced by the computer model were appropriate in the given context and were linguistically correct which means the model is able to perform well across different images.

True: ગુલાબી ડ્રેસ પહેરેલી સ્ત્રી તાજમહેલના ભવ્ય દૃશ્યની સામે સુંદર રીતે પોઝ આપે છે.
Generated: એક સ્ત્રી અગરબૂમમાં શાંત વાદળી પાણીના પૂલ સાથે તાજમહેલની સામે પોઝ આપે છે.



True: ગુજરાતમાં એક રંગીન પતંગ મહોત્સવ છે જેમાં ઘણા લોકો આ પ્રદર્શનનો આનંદ માણી રહ્યા છે.
Generated: ગુજરાતમાં પતંગ ઉડાવતા ઉત્સવમાં મોટી સંખ્યામાં લોકો ભેગા થાય છે.



Figure 7 - Inference Chart

4. API Implementation

To facilitate practical use of the model, an API endpoint was created with FastAPI. The image API was designed to accept inputs in the form of images and return outputs in the form of caption in Gujarati which makes it easy to integrate with other systems. The API was also Dockerized to make it easy to deploy in different environments.

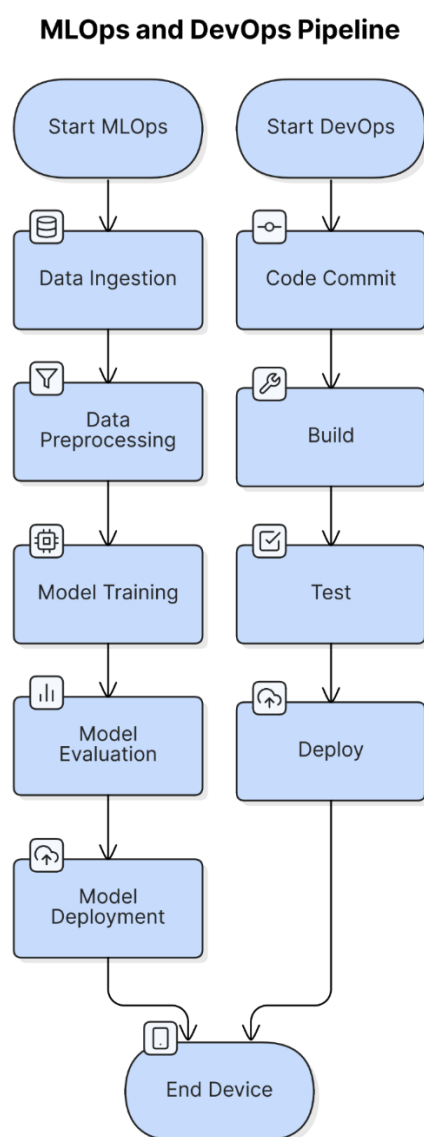


Figure 8 - Development Pipeline

5. Mobile Application

To enhance the practicality of the model, a mobile application was created with the help of React Native framework building tools using Expo technology. The application has a straightforward and user-friendly interface that enables users to upload images and obtain captions from their devices. This approach which is easy to use assists the end users in getting the advantages of the project irrespective of the level of computer literacy.

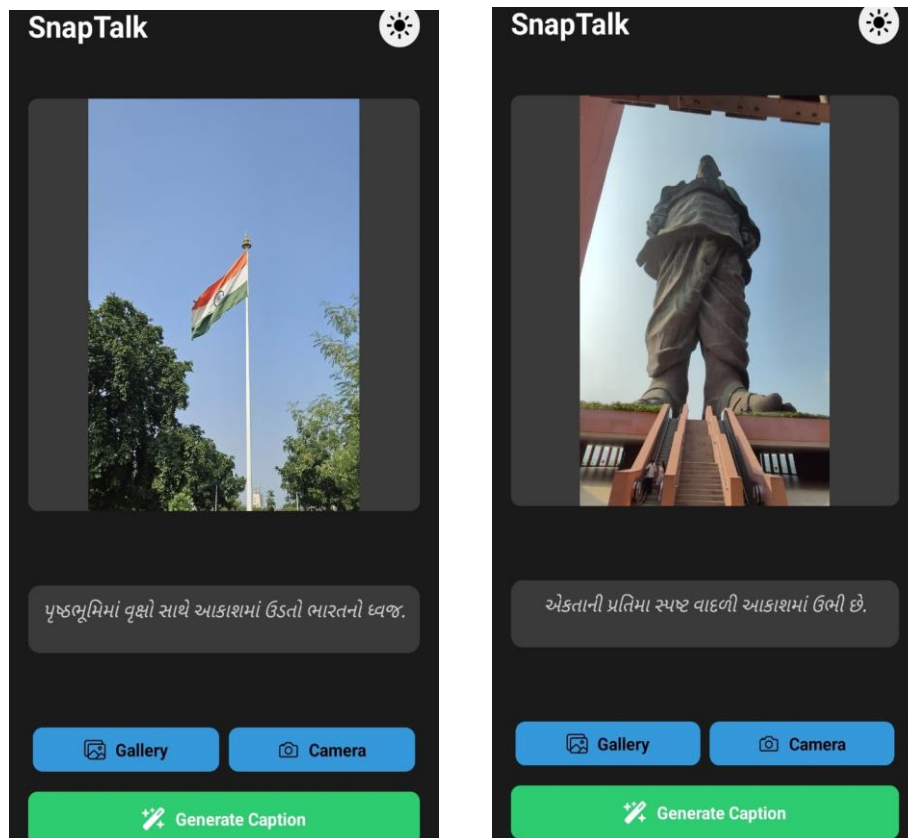


Figure 9 - Mobile Application

Conclusion

The Snaptalk project managed to demonstrate the effective use of sophisticated vision-language models in image captioning for Gujarati, which is a low-resourced language. The project was able to fill in the large gaps by producing captions that were both linguistically and culturally contextualized via the fine-tuned BLIP-Large model. The training of the model to generate contextual captions was also facilitated by a well-designed dataset of around 1800 images representing the diverse cultural settings of Gujarat. Fine-tuning maintained the generalization ability of the model and avoided problems of catastrophic forgetting, being able to adapt to specific tasks within that domain [1][8].

It is also worth mentioning that the theoretical progress made in this project did not remain merely on paper as they also focused on developing a working application. This app was designed to be simple, allowing users to upload an image and obtain a caption created by the fine-tuned BLIP-Large model. The development of this app on cloud platforms ensured its scalability, robustness and accessibility. The picture processing, the caption creation and the notification of the accomplishment of the tasks were carried out using the Flask based API seamlessly and thus responded to the requirements of the technology and end-users activities [3].

The model's effectiveness in producing captions that not only met but also exceeded linguistic and cultural expectations was confirmed by two evaluation metrics, BLEU and ROUGE. This outcome demonstrates the importance of merging high-quality datasets, cutting-edge AI models, and user-focused implementation for overcoming the digital divide in languages that are less resourceful.

Powers of AI-embedded solutions imbuing Gujarati have enabled Snaptalk to not only foster digital inclusion but also led the way for other under-resourced languages. This work emphasizes how AI can be used to promote cultural inclusivity, diversity, and preservation in the approach of technology. It guarantees that there is a post-research benefit to this project and a pragmatic tool available to users while also ensuring future opportunities [12][15].

Scope of Future Work

1. Text-to-Voice Feature

The proposed improvements include the addition of a text-to-voice feature in the mobile app. Users who are visually impaired or illiterate will benefit from hearing the captions, which will be generated in Gujarati, read aloud through this text-to-speech technology. This would also make the application more appealing and efficient since it would utilize available Gujarati TTS voice overs.

2. User Feedback Feature

To enhance the precision and relevance of the captions being generated, a user verification feature is going to be added. Users will be able to comment on the captions as follows:

- Anyone can comment on the caption that is always wrong and suggest an alternative.
- This data will be available for the next round of continual learning in the system and will be focused on the problem of users addressing their feedback.

When users become researchers, the model is assumed to self-learn and be improved upon during its lifetime.

3. Improved Dataset and Annotation Tools

Future work could involve expanding the dataset by sourcing more culturally diverse images and developing semi-automated annotation tools. These tools could streamline the process of creating high-quality, domain-specific datasets, which are critical for further fine-tuning the model.

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